

EXP8: Depth First Search (DFS)

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Aim:

To develop a Python program that implements the Depth First Search (DFS) algorithm to traverse a graph by exploring as far as possible along each branch before backtracking.

Algorithm:

Step 1: Represent the graph using an adjacency list (dictionary of lists).

Step 2: Create an empty stack (or use recursion) and push the starting node onto it.

Step 3: Pop a node from the stack; if it is not visited, mark it as visited and add it to the traversal result.

Step 4: Push all unvisited neighbours of the current node onto the stack.

Step 5: Repeat Steps 3 and 4 until the stack becomes empty.

Step 6: Display the DFS traversal order.

Source Code:

```
# Program to implement Depth First Search (DFS)
```

```
def dfs(graph, start, visited=None, order=None):
```

```
    if visited is None:
```

```
        visited = set()
```

```
    if order is None:
```

```
        order = []
```

```
    visited.add(start)
```

```
    order.append(start)
```

```
    for neighbour in graph[start]:
```

```
        if neighbour not in visited:
```

```
            dfs(graph, neighbour, visited, order)
```

```
    return order
```

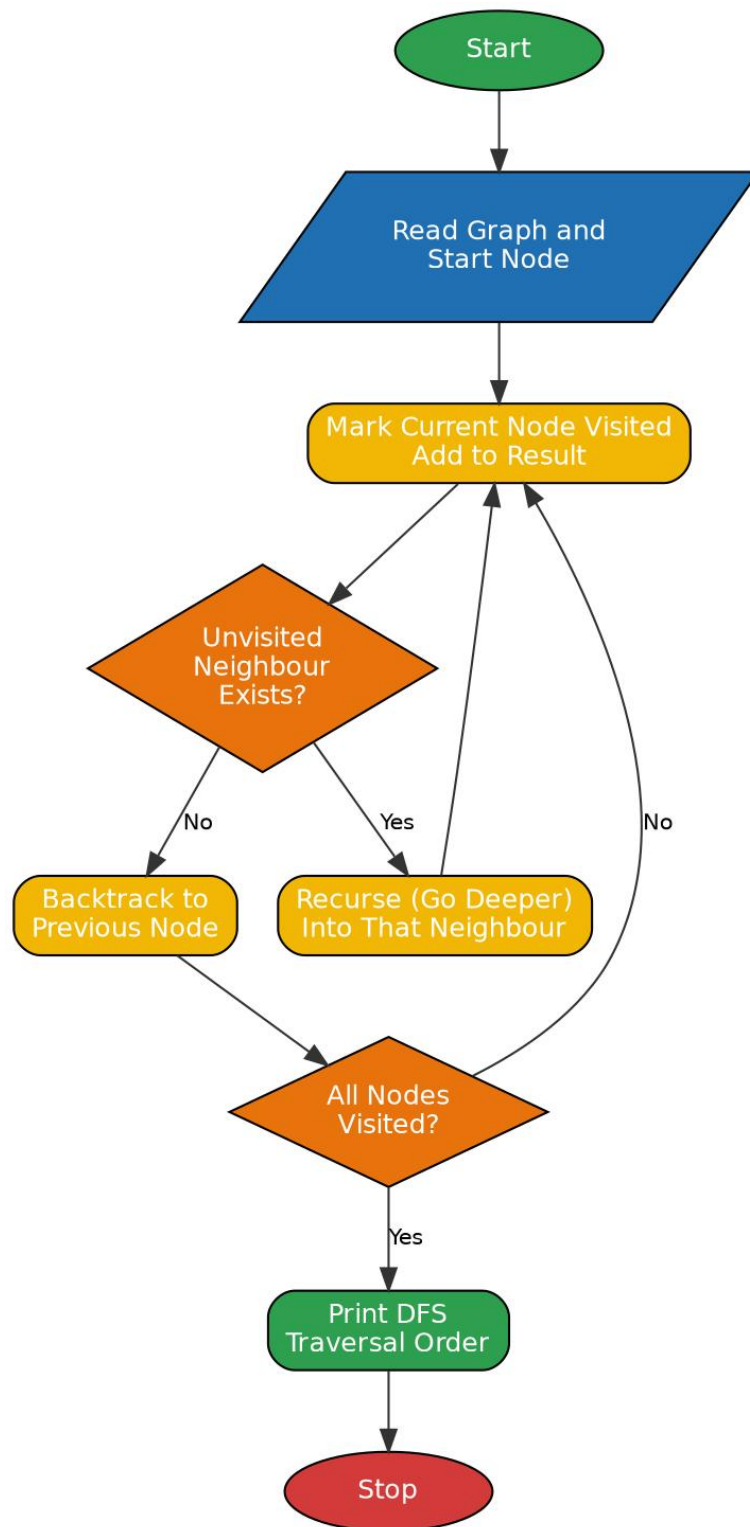
```
graph = {  
    'A': ['B', 'C'],  
    'B': ['A', 'D', 'E'],  
    'C': ['A', 'F'],  
    'D': ['B'],  
    'E': ['B', 'F'],  
    'F': ['C', 'E']  
}
```

```
start_node = 'A'
```

```
result = dfs(graph, start_node)
```

```
print("DFS Traversal starting from", start_node, ":")  
print("-> ".join(result))
```

Flowchart:



Output:

```
>>> ===== RESTART: C:/Users/yanar/Documents/CSA1717/EXP8.py =====  
DFS Traversal starting from A :  
A -> B -> D -> E -> F -> C  
>>> |
```

Result:

Thus the program to implement Depth First Search (DFS) was written, executed successfully and the output was verified.